

LYDOCAN INJECTION 2%

Ingredient(s):

Each ml contains:

Lidocaine Hydrochloride 20mg
Methyl Paraben (as preservative) 1mg

Pharmacodynamic:

1. Lidocaine reversibly blocks impulse conduction in the nerve fibres by inhibiting the transport or preventing the inward movement of sodium ions through the nerve membrane. Local anaesthetics of the amide type are thought to act within the sodium channels of the nerve membrane.
2. Lidocaine has a rapid onset of effect, a high anaesthetic frequency and low toxicity. Lower concentrations of lidocaine have less effect on motor nerve fibres.
3. Local anaesthetic drugs may have similar effects on excitatory membranes in the brain and myocardium.

Pharmacokinetic:

1. Lidocaine has a rapid onset and a medium duration of action. The onset of action is 1-5 min following infiltration and 5-15 min following other types of administration.
2. The rate of absorption is dependent on the dose, route of administration and vascularity of or perfusion at the injection site. Intercostal blockades produce the highest peak plasma concentrations (approximately 1.5 mcg/mL per 100 mg injected), while abdominal SC injections produce the lowest plasma concentrations (approximately 0.5 mcg/mL per 100 mg injected). Epidural and major nerve block produce peak plasma levels intermediate between these.
3. The plasma binding of lidocaine is dependent on drug concentration and the fraction bound decreases with increasing concentration. At concentrations of 1-4 mcg/mL of free base, 60-80% of lidocaine is protein bound. Binding is also dependent on the plasma concentration of the α 1-acid glycoprotein (65%).
4. Lidocaine crosses the blood-brain and placental barriers by passive diffusion. Since the degree of plasma protein-binding in the fetus is less than in the mother, although free lidocaine concentrations will be the same, the total plasma concentration will be greater in the mother.
5. Lidocaine is eliminated predominantly through metabolism. Lidocaine has a total plasma clearance of 0.95 L/min. Less than 10% of a dose is excreted unchanged in the urine.
6. The duration of action depends upon the concentration used, the dose given, the nerves to be blocked and the status of the patient. The 2% solution will produce an effect up to 5 hrs when given as a peripheral nerve block.
7. The rate of metabolism of lidocaine appears to be limited by liver blood flow, which may be reduced in patients after acute myocardial infarction and/or congestive heart failure. The rate of lidocaine metabolism may also be reduced in patients with liver or hepatic tissue necrosis, possibly because of altered perfusion.
8. On account of the rapid hepatic metabolism, the kinetics are sensitive to all alterations in liver function. The half-life can be more than doubled in patients with impaired liver function. Impaired renal function does not affect the kinetics, but can increase the accumulation of metabolites.

Indication:

To be used for most major nerve blocks eg, sciatic and retrobulbar.

Dosage and Administration:

Adults and Children >12 years:

This product should only be used by physicians with experience of regional anaesthesia or under their supervision. The lowest possible dose for adequate anaesthesia should be used.

For parenteral use.

Maximum Recommended Doses: 20 mg/mL, 20 mL (lidocaine HCl 400 mg).

Table 1 illustrates the dosage recommendations, which serves as a guide to dosage for the more commonly used techniques in the average adult. The clinician's experience and knowledge of the patient's physical status are of importance in calculating the required dose. When prolonged blocks are used eg, by repeated administration, the risks of reaching a toxic plasma concentration or inducing a local neural injury must be considered (see Table 1).

Table 1. Dosage Recommendations.					
Type of Block	Concentration (mg/mL)	Dose		Onset (min)	Duration (hr)
		mL	mg		
Major nerve block					
Retrobulbar	20	4	80	3-5	1.5-2
Sciatic	20	15-20	300-400	15-30	2-3

The doses given in Table 1 are those considered to be necessary to produce successful blocks and should be regarded as a guide for use in adults. Individual variations in onset and duration occur. The figures reflect the expected average dose range needed. Standard textbooks should be consulted for factors affecting specific block techniques and for individual patient requirements.

Unnecessarily high doses of local anaesthetics are to be avoided. In general, surgical

anaesthesia requires the use of the higher concentrations and doses. When blocking smaller nerves or when a less intense block is required, the use of a lower concentration is indicated. The volume of drug used will affect the extent and spread of anaesthesia.

In order to avoid intravascular injection, aspiration should be repeated prior to and during administration of the main dose, which should be injected slowly or in incremental doses at a rate of 100-200 mg/min while closely observing the patient's vital functions and maintaining verbal contact. An inadvertent intravascular injection may be recognised by a temporary increase in heart rate and an accidental intrathecal injection by signs of a spinal block. If toxic symptoms occur, the injection should be stopped immediately.

The doses should be reduced for children and patients in reduced general condition.

Contraindication(s):

1. Hypersensitivity to amide-type local anaesthetics.
2. This product should not be used in patients who are hypersensitive to methylparaben or to its metabolite, paraaminobenzoic acid (PABA).
3. This product should be avoided in patients allergic to ester-type local anaesthetics that also has the metabolite PABA.
4. Local anaesthetic techniques must not be used when there is inflammation and/or sepsis in the region of the proposed injection and/or in the presence of septicaemia.
5. Patients with Stokes-Adam syndrome, Wolff-Parkinson-White syndrome or with severe degrees of sinoatrial, atrioventricular or intraventricular block in the absence of an artificial pacemaker.

Precaution(s) / Warning(s):

1. When any local anaesthetic agent is used, resuscitative equipment and drugs including oxygen, should be immediately available in order to manage possible adverse reactions involving the cardiovascular, respiratory or CNS. Because of the possibility of hypotension and bradycardia following major blocks, an IV cannula should be inserted before the local anaesthetic is injected. Delay in proper management of dose-related toxicity, under-ventilation from any cause and/or altered sensitivity may lead to the development of acidosis, cardiac arrest and death. Like all local anaesthetic agents, lidocaine can cause acute central nervous and cardiovascular toxic effects in cases of use that leads to high concentrations in the blood. This applies particularly after inadvertent IV administration.
2. Injection should always be made slowly with frequent aspirations to avoid inadvertent intravascular injection which can produce cerebral symptoms even at low doses.
3. Careful and constant monitoring of cardiovascular and respiratory vital signs and the patient's state of consciousness should be accomplished after each local anaesthetic injection. It should be kept in mind that at such times, restlessness, anxiety, tinnitus, dizziness, blurred vision, tremors, depression or drowsiness may be early warning signs of CNS toxicity.
4. Caution should be exercised in patients of the following categories: Elderly and patients with generally reduced general condition; patients with II or III degree AV block since local anaesthetics can lower the conduction capacity of the myocardium; patients with severe hepatic disease or severely impaired renal function; patients treated with class III antiarrhythmic drugs (eg, amiodarone) should be closely observed and electrocardiogram (ECG) monitoring should be considered, since the cardiac effects of lidocaine and class III antiarrhythmic drugs can be additive. (See Interaction with other medicaments)
5. The safety and effectiveness of this product depends on proper dosage, correct technique and adequate precautions. Standard textbooks should be consulted regarding specific techniques and precautions for various regional anaesthetic procedures.
6. The lowest dosage that results in effective anaesthesia should be used (see Dosage and Administration). Repeated injection of this product may cause accumulation of lidocaine or its metabolites and result in toxic effects.
7. Tolerance to elevated blood levels varies with the status of the patient. Elderly, young or debilitated patients, including those with advanced liver disease or severe renal dysfunction, should be given reduced doses commensurate with their age and physical condition.
8. Lidocaine should be given with great caution to patients with epilepsy, impaired cardiac conduction, bradycardia, severe shock or digitalis intoxication. Lidocaine should also be administered with great caution to patients with impaired cardiovascular function as they may be less able to compensate for functional changes associated with the prolongation of AV conduction produced by these drugs. In patients with Stokes-Adams or Wolff-Parkinson-White syndrome, extreme care should be taken to avoid accidental arterio-venous injection.
9. Since lidocaine is metabolised in the liver and excreted via the kidneys, the possibility of drug accumulation should be considered in patients with hepatic and/or renal impairment.
10. Inadvertent intravascular or subarachnoid injection of small doses of local anaesthetics injected into the head and neck area, including retrobulbar, dental and stellate ganglion blocks, may produce adverse reactions similar to systemic toxicity seen with unintentional intravascular injections of larger doses.
11. Clinicians who perform retrobulbar blocks should be aware that there have been reports of cardiovascular collapse and apnoea following the use of local anaesthetic injections for retrobulbar block. Prior to retrobulbar block, necessary equipment, drugs and personnel should be immediately available as with all other

- regional procedures. Retrobulbar injections can, in rare cases, reach the subarachnoid space, causing temporary blindness, cardiovascular collapse, apnoea, convulsions, etc. These must be diagnosed and treated promptly.
12. Retro- and peribulbar injections of local anaesthetics carry a low risk of persistent ocular muscle dysfunction. The primary causes include traumatic nerve injury and/or local toxic effects on muscles or nerves. The severity of such tissue damage is related to the degree of trauma, the concentration of the local anaesthetic and the duration of exposure of the tissue to the local anaesthetic. For this reason, as with all local anaesthetics, the lowest effective concentration and dose of local anaesthetic should be used. Vasoconstrictors may aggravate tissue reactions and should be used only when indicated.
 13. Fetal bradycardia/tachycardia frequently follows paracervical block and may be associated with fetal acidosis and hypoxia. Occasional cases of perinatal morbidity and mortality have been reported. When the recommended dose is exceeded, the risk of fetal bradycardia increases. Careful monitoring of the fetal heart rate is necessary.
 14. Lidocaine should be used with caution in patients with known drug sensitivities.
 15. This product is probably porphyrinogenic and should only be prescribed to patients with acute porphyria if there are strong or urgent indications. Appropriate precautions should be taken for all porphyric patients.
 16. Effects on the Ability to Drive or Operate Machinery: Depending on the dose and method of administration, local anaesthetics may have a very mild effect on mental function and may temporarily impair locomotion and coordination.

Interaction with Other Medicaments:

1. Drugs that inhibit the metabolism of lidocaine (eg, cimetidine) may cause potentially toxic plasma concentrations when lidocaine is given in repeated high doses over a long period of time.
2. Lidocaine should be used with caution with other local anaesthetics or class IB antiarrhythmic drugs, as the toxic effects are additive.
3. No specific interaction studies with local anaesthetics and class III antiarrhythmic drugs (eg, amiodarone) have been carried out, but caution is recommended.
4. Antiarrhythmic Drugs: Local anaesthetics of the amide type eg, lidocaine should be used with caution in patients receiving other local anaesthetics or agents structurally related to amide-type local anaesthetics ie, certain antiarrhythmic drugs eg, disopyramide, procainamide, mexilitene since potentiation of cardiac effects may occur. Specific interaction studies with lidocaine and class III antiarrhythmic drugs (eg, amiodarone) have not been performed, but caution should be advised.
5. Amiodarone: Amiodarone has been reported to reduce the clearance of lidocaine. This combination has been reported to precipitate seizures and to lead to severe sinus bradycardia and a long sinoatrial arrest. Patients receiving the combination should be monitored carefully.
6. Beta-Adrenoreceptor Antagonists: Propranolol and metoprolol reduce the metabolism of IV administered lidocaine. If these drugs are administered concurrently, the patient should be closely observed for signs of lidocaine toxicity.
7. Cimetidine: Cimetidine reduces the clearance of IV administered lidocaine and toxic effects due to high serum lidocaine levels have been reported when these 2 drugs have been administered concurrently.
8. Anticonvulsive Agents: Phenytoin and other antiepileptic drugs eg, phenobarbitone, primidone and carbamazepine appear to enhance the metabolism of lidocaine. Phenytoin and lidocaine have additive cardiac depressant effects.
9. Inhalational Anaesthetics: Lidocaine decreases the minimum effective concentration of inhalational anaesthetics eg, nitrous oxide.
10. Skeletal Muscle Relaxants: Lidocaine and skeletal muscle relaxants eg, suxamethonium, lead to excessive neuromuscular blockade; therefore, this combination must be used with caution.
11. Structurally-Related Local Anaesthetics: Lidocaine should be used with caution in patients receiving agents structurally related to local anaesthetics.
12. Creatine Kinase: The IM injection of lidocaine may result in an increase in creatine kinase levels for up to 48 hrs. This may interfere with the diagnosis of myocardial infarction.
13. Incompatibilities: Alkalinisation can cause precipitation as lidocaine is limitedly soluble at pH >6.5.

Pregnancy and Lactation:

Use in pregnancy:

The safe use of lidocaine during pregnancy has not been established. Lidocaine passes the placenta. Although lidocaine has been used extensively for surgical procedures during pregnancy with no reports of ill effects to mother or fetus, there are no adequate or well-controlled studies in pregnant women of the effect of lidocaine on the developing fetus.

At temporary use during pregnancy and labour, the benefit is judged to outweigh the possible risks. Paracervical blockade or pudendal blockade with lidocaine increases the risk for reactions of the foetus eg, bradycardia/tachycardia. Therefore, the heart rate of the foetus must be carefully monitored.

Use in lactation:

Small amounts of Lidocaine are secreted into breast milk and the possibility of an

allergic reaction in the infant, albeit remote, should be borne in mind when using lidocaine in nursing mothers.

Side Effect(s):

Undesirable effects caused by Lidocaine itself can be difficult to distinguish from the physiological effects of the nerve block (eg, fall in blood pressure, bradycardia), events caused directly by the needle puncture (eg, nerve damage) or caused indirectly by the needle puncture.

1. Gastrointestinal Disorders: Nausea, vomiting.
2. Vascular Disorders: Hypotension, hypertension.
3. Cardiac Disorders: Bradycardia, cardiac arrest and arrhythmias.
4. Nervous System Disorders: Paraesthesia, dizziness, symptoms of CNS toxicity (convulsions, circumoral paraesthesia, numbness of the tongue, hyperacusis, visual disturbances, unconsciousness, tremor, lightheadedness, tinnitus, dysarthria), neuropathy, peripheral nerve injury, arachnoiditis.
5. Immune System Disorders: Allergic reactions, in the most severe cases, anaphylactic shock.
6. Eyes: Double vision.
7. Respiratory, Thoracic and Mediastinal Disorders: Respiratory depression.

Symptoms and Treatment for Overdosage, and Antidote(s):

Accidental intravascular injections of local anaesthetics can cause immediate system-toxic reactions (within seconds to a few minutes). Signs of system toxicity in cases of overdose occur later (15-60 min after injection) due to a slower increase in the concentration of local anaesthetic in the blood.

50 mg IV to a 1-month old child, 200-400 mg infiltration to an adult, 500 mg to an 80-year old patient or 1 g IV to adults can cause very severe intoxication.

Symptoms

First CNS excitation, later CNS depression. When exposed for large doses, the first symptom can be rapidly developing seizures. Agitation, dizziness, visual disturbances, perioral paraesthesia, nausea. Later, ataxia, auditory disturbances, euphoria, confusion, speech difficulties, paleness, sweating, tremor, seizures, coma, apnoea. Different kinds of arrhythmias especially bradycardial arrhythmias but also, if exposed to large doses, ventricular tachycardia, ventricular fibrillation, QRS prolongation, AV block. Cardiac failure, hypotension (rare cases of methaemoglobinaemia are described).

Treatment

Seizures are treated with diazepam. Oxygen. If needed, tracheal intubation and controlled respiration (if needed hyperventilation). Bradycardia is treated with atropine. Circulatory deterioration is treated with fluids administered IV, dobutamine and if needed, epinephrine (initially 0.05 mcg/kg body weight/min, to be increased if needed by 0.05 mcg/kg body weight/min each 10 min), in more severe cases, guided by haemodynamic monitoring. Ephedrine may also be tried. If circulatory arrest, resuscitation efforts for several hours may be indicated.

Incompatibilities:

Lidocaine causes precipitation of amphotericin, methohexitone sodium and sulphadiazine sodium in glucose injection. It is recommended that admixtures of lidocaine and glyceryl trinitrate should be avoided.

Shelf-Life:

3 years from the date of manufacture.

Storage Condition(s):

Store at temperature below 30°C.

Product Description & Packing(s):

A clear and colorless solution.

10ml x 10 & 10ml x 20 vials.



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